UNIT II : Convection

Topics: convection basics, h, free & forced convection, boundary layers, empirical relations, Reynolds–Colburn analogy, integral solution, internal flow

2.1 Basic Concepts of Convection

Convection is heat transfer between a solid surface and a moving fluid. It combines conduction (molecular diffusion at the wall) and advection (bulk motion of fluid).

Newton's law of cooling gives the rate:

$$Q = h\,A\,(T_s - T_\infty)\quad q = h\,(T_s - T_\infty)$$ Newton

→ Local coefficient h_x varies along the surface; average $h = (1/L)\int h_x\,dx$

→ At the wall fluid velocity = 0 (no-slip), so heat leaves by pure conduction: $h = -k_f\,(\partial T/\partial y)_{y=0} / (T_s - T_\infty)$

→ h is NOT a property – it depends on velocity V, ρ, μ, c_p, k, geometry L, surface roughness, flow type

Types of convection

Type	Cause of motion	Example	h (W/m^2K)
Forced	fan, pump, wind	radiator, heat exchanger	25 – 250 (air), 50 – 20000 (water)
Natural (free)	density difference (buoyancy)	hot plate in still air	2 – 25 (air)
Mixed	both comparable ($Gr/Re^2 \approx 1$)	slow flow in heated pipe	—

→ Also : laminar vs turbulent; external (plate, cylinder) vs internal (pipe, duct); with or without phase change

Dimensionless numbers

Number	Expression	Physical meaning
Reynolds Re	$\rho V L/\mu = VL/\nu$	inertia force / viscous force
Prandtl Pr	$\mu c_p/k = \nu/\alpha$	momentum diffusivity / thermal diffusivity
Nusselt Nu	hL/k	convection / conduction in fluid layer
Grashof Gr	$g\beta\Delta T L^3/\nu^2$	buoyancy force / viscous force
Rayleigh Ra	$Gr \cdot Pr$	free convection driving strength
Stanton St	$h/\rho V c_p = Nu/(Re\ Pr)$	heat transferred / heat capacity of flow
Peclet Pe	$Re \cdot Pr$	advection / diffusion of heat

Dimensional analysis (Buckingham π): Forced : $Nu = f(Re, Pr)$ Free : $Nu = f(Gr, Pr) = f(Ra, Pr)$ (usually $Nu = C\ Re^m\ Pr^n$).

Note: Fluid properties are evaluated at the film temperature $T_f = (T_s + T_\infty)/2$ (unless stated otherwise). For gases, volume expansion coefficient $\beta = 1/T_f$ (Kelvin).

Dimensional analysis – why Nu = f(Re, Pr)

Forced convection depends on 7 variables : $h, \rho, V, L, \mu, c_p, k$. Fundamental dimensions $M, L, T, \theta = 4$. By Buckingham π-theorem number of dimensionless groups = 7 - 4 = 3. Choosing repeating variables (ρ, V, L, k) :

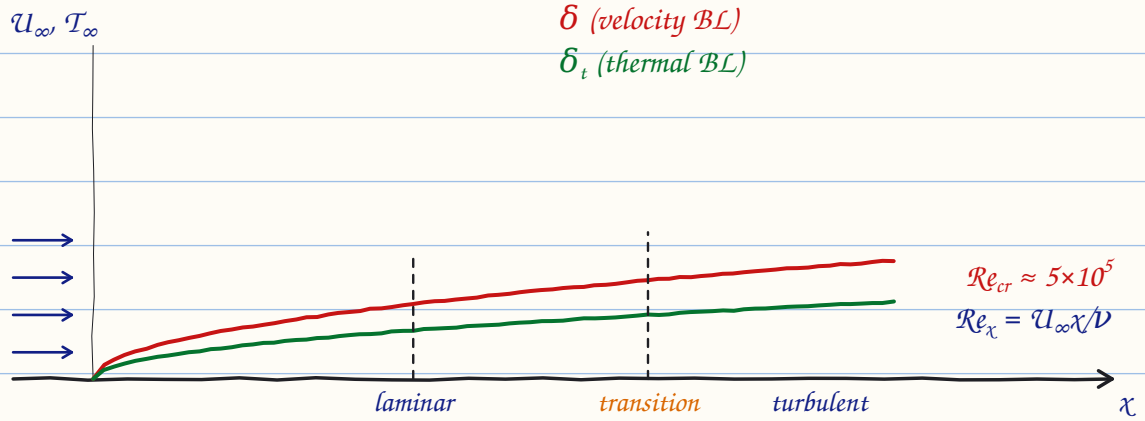
$$\pi_1 = hL/k = Nu\ ,\ \pi_2 = \rho VL/\mu = Re\ ,\ \pi_3 = \mu c_p/k = Pr$$

Hence $Nu = f(Re, Pr)$. For natural convection V is replaced by the buoyancy term $g\beta\Delta T$ and the group $(\rho^2 g\beta\Delta T L^3/\mu^2) = Gr$ appears, so $Nu = f(Gr, Pr)$.

→ The experimental constants C, m, n are then found by plotting log Nu against log Re (or log Ra): a straight line of slope m

Boundary Layer Concepts (External Flow)

Uniform stream U_∞ meets a flat plate at temperature T_s. Fluid sticks to the wall; velocity rises from 0 to U_∞ over a thin region:



→ *Hydrodynamic BL thickness* δ : *distance from wall where* $u = 0.99\,U_\infty$; *wall shear* $\tau_w = \mu\,(\partial u/\partial y)_0$, $C_f = 2\tau_w/\rho U_\infty^2$

→ *Thermal BL thickness* δ_t : *where* $(T_s - T)/(T_s - T_\infty) = 0.99$; *all heat transfer happens inside* δ_t

→ *Ratio :* $\delta_t/\delta \approx Pr^{-1/3}$ → *liquid metals* $(Pr \ll 1)\ \delta_t \gg \delta$; *oils* $(Pr \gg 1)\ \delta_t \ll \delta$; *gases* $Pr \approx 0.7 \rightarrow \delta \approx \delta_t$

→ *Laminar → turbulent at* $Re_{cr} \approx 5\times10^5$ *for a flat plate; turbulent BL is thicker, gives larger h and larger drag*

Boundary-layer equations (steady, 2-D, laminar, constant properties)

$$\partial u/\partial x + \partial v/\partial y = 0 \text{ (continuity)}$$

$$u\,\partial u/\partial x + v\,\partial u/\partial y = \nu\,\partial^2 u/\partial y^2 \text{ (momentum)}$$

$$u\,\partial T/\partial x + v\,\partial T/\partial y = \alpha\,\partial^2 T/\partial y^2 \text{ (energy)}$$

Similarity (Blasius) solution of these gives the exact laminar results quoted below. When $Pr = 1$ *the momentum and energy equations are identical in form, so velocity and temperature profiles coincide.*

Approximate (Integral) Solution – Laminar BL

Von Karman momentum integral equation for a flat plate $(dp/dx = 0)$:

$$\tau_w = \rho\ d/dx \int_0^\delta u\,(U_\infty - u)\,dy$$

Assume a cubic velocity profile satisfying $u = 0$ *at* $y = 0$, $u = U_\infty$ *and* $\partial u/\partial y = 0$ *at* $y = \delta$:

$$u/U_\infty = (3/2)(y/\delta) - (1/2)(y/\delta)^3$$

Then $\tau_w = 3\mu U_\infty/2\delta$ *and* $\int u(U_\infty - u)dy = (39/280)\,U_\infty^2\,\delta$. *Substituting :* $(39/280)$

$\rho U_\infty^2 \, d\delta/dx = 3\mu U_\infty/2\delta \rightarrow \delta \, d\delta = (140/13)(\nu/U_\infty)\, dx$. Integrating with $\delta(0) = 0$:

$$\delta_x = 4.64\, x/\sqrt{Re_x}$$
$$C_{fx} = 0.646/\sqrt{Re_x} \text{ (exact: } 0.664/\sqrt{Re_x})$$
$$C_{f,avg} = 1.328/\sqrt{Re_L}$$

Thermal integral equation: $d/dx \int_0^{\delta_t} (T_\infty - T)\, u\, dy = \alpha\, (\partial T/\partial y)_0$. With a cubic temperature profile $\theta = (3/2)(y/\delta_t) - (1/2)(y/\delta_t)^3$ ($\theta = (T - T_s)/(T_\infty - T_s)$) and the above $u(y)$:

$$\delta_t/\delta = Pr^{-1/3}/1.026$$
$$Nu_x = h_x x/k = 0.332\, Re_x^{1/2}\, Pr^{1/3}$$
$$Nu_L = 0.664\, Re_L^{1/2}\, Pr^{1/3} \quad (h_{avg} = 2h_L)$$

Note: Valid for $Pr \geq 0.6$, laminar flow ($Re_x < 5\times10^5$), constant wall temperature, properties at film temperature. For constant heat flux surface : $Nu_x = 0.453\, Re_x^{1/2}\, Pr^{1/3}$.

Turbulent flat plate – 1/7 power-law result

Take $u/U_\infty = (y/\delta)^{1/7}$ and the pipe-flow wall shear $\tau_w = 0.0225\, \rho U_\infty^2 (\nu/U_\infty\delta)^{1/4}$. Then $\int u(U_\infty - u)dy = (7/72)U_\infty^2\delta$. The momentum integral gives:

$$\delta = 0.37\, x\, Re_x^{-1/5}$$
$$C_{f,x} \approx 0.0592\, Re_x^{-1/5}$$
$$C_{f,avg} = 0.074\, Re_L^{-1/5} \quad (5\times10^5 < Re_L < 10^7)$$

→ Turbulent δ grows as $x^{0.8}$ (faster than laminar $x^{0.5}$); eddies mix the fluid so wall gradients and h are much larger

→ Using Colburn analogy: $Nu_x = (C_{f,x}/2)\, Re_x\, Pr^{1/3} = 0.0296\, Re_x^{0.8}\, Pr^{1/3}$

Fluid Friction – Heat Transfer Analogy

Momentum and heat transfer in a boundary layer are governed by similar equations, so skin friction and heat transfer coefficient are related:

Analogy	Relation	Valid for
Reynolds	$St = C_f/2$	$Pr = 1$ (laminar & turbulent)
Prandtl–Taylor	$St = (C_f/2) / [1 + 5\sqrt{(C_f/2)}(Pr-1)]$	$Pr \neq 1$
Chilton–Colburn	$St \cdot Pr^{2/3} = C_f/2$ ($j_H = j_f$)	$0.6 < Pr < 60$

→ Flat plate : $C_{f,x} = 0.664\ Re_x^{-1/2}$ laminar ; $0.0592\ Re_x^{-1/5}$ turbulent. Hence knowing friction, h is found: $h = (C_f/2)\ \rho\ c_p\ U_\infty\ Pr^{-2/3}$

→ Practical use : drag measured in wind tunnel → heat transfer estimated (and vice-versa); explains why good heat-transfer surfaces create high pressure drop

Reynolds analogy – quick derivation

Shear stress: $\tau = (\mu + \rho\varepsilon_m)\ du/dy$; heat flux: $q = -(k + \rho c_p \varepsilon_h)\ dT/dy$. If $Pr = 1$ and $\varepsilon_m = \varepsilon_h$, then $q/\tau = -c_p\ dT/du$. Integrating from the wall to the free stream ($\tau \approx \tau_w$, $q \approx q_w$):

$$q_w/\tau_w = c_p(T_s - T_\infty)/U_\infty \rightarrow h/(\rho\ c_p\ U_\infty) = C_f/2 \text{ i.e. } St = C_f/2$$

Colburn extended this to $0.6 < Pr < 60$ by introducing $Pr^{2/3}$. Used chiefly for turbulent flow and for estimating h from a measured pressure drop in pipes.

Forced Convection – External Flow Correlations

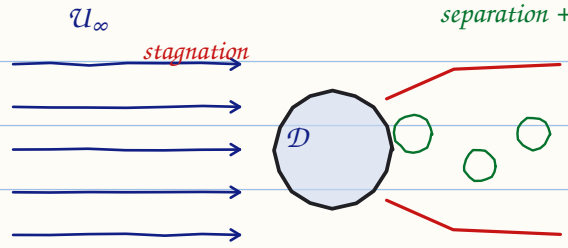
(a) Flat plate (properties at T_f)

Flow condition	Local Nu_x	Average Nu_L
Laminar, $Re < 5\times10^5$, $Pr>0.6$	$0.332\ Re_x^{0.5}\ Pr^{1/3}$	$0.664\ Re_L^{0.5}\ Pr^{1/3}$
Turbulent, $Re > 5\times10^5$	$0.0296\ Re_x^{0.8}\ Pr^{1/3}$	$0.037\ Re_L^{0.8}\ Pr^{1/3}$
Mixed (laminar + turbulent)	—	$(0.037\ Re_L^{0.8} - 871)\ Pr^{1/3}$
Liquid metals, $Pr < 0.05$	$0.565\ (Re_x\ Pr)^{0.5}$	$1.13\ (Re_L Pr)^{0.5}$

→ Constant heat flux : $Nu_x = 0.453\ Re_x^{1/2}\ Pr^{1/3}$ (laminar), $Nu_x = 0.0308\ Re_x^{0.8}\ Pr^{1/3}$ (turbulent)

→ Heat transfer rate $Q = h_{avg}\ A\ (T_s - T_\infty)$; drag $F = C_{f,avg}\ (\frac{1}{2}\rho U_\infty^2)\ A$

(b) Cylinder in cross-flow ($Re_D = U_\infty D/\nu$)



Laminar BL : separation at ≈ 80°

Turbulent BL : at ≈ 140° ($Re > 2\times10^5$)

h is max at the front stagnation point, min near separation.

Hilpert : $Nu_D = C\,Re_D^{\,n}\,Pr^{1/3}$

Re_D	0.4–4	4–40	40–4000	4000–40000	40000–400000
C	0.989	0.911	0.683	0.193	0.027
n	0.330	0.385	0.466	0.618	0.805

Churchill–Bernstein (all $Re\cdot Pr > 0.2$): $Nu_D = 0.3 + 0.62\,Re_D^{1/2}Pr^{1/3} / [1 + (0.4/Pr)^{2/3}]^{1/4} \cdot [1 + (Re_D/282000)^{5/8}]^{4/5}$

→ Sphere (Whitaker) : $Nu_D = 2 + (0.4\,Re_D^{1/2} + 0.06\,Re_D^{2/3})\,Pr^{0.4}\,(\mu_\infty/\mu_s)^{1/4}$; $Nu \to 2$ for still fluid

→ Non-circular cylinders / tube banks use the same form with different C, n (Jakob, Zhukauskas); in banks velocity is taken at minimum flow area

Solved Example 1 – Flat plate, laminar

Air at 20°C flows at 5 m/s over a 1 m long, 1 m wide plate at 80°C. $T_f = 50°C$: $\nu = 1.79\times10^{-5}$, $k = 0.028$ W/mK, $Pr = 0.703$. Find average h and heat loss.

$Re_L = 5(1)/1.79\times10^{-5} = 2.79\times10^5 < 5\times10^5$ → laminar. $Nu_L = 0.664$

$(2.79\times10^5)^{0.5}(0.703)^{1/3} = 0.664 \times 528.5 \times 0.889 = 312$

$h = Nu\,k/L = 312 \times 0.028 / 1 =$ **8.7 W/m²K**; $Q = hA\Delta T = 8.7 \times 1 \times 60 =$ **524 W** (one side)

Solved Example 2 – Cylinder in cross-flow

Air at 20°C, 10 m/s over a 25 mm tube at 120°C. At $T_f = 70°C$: $\nu = 1.995\times10^{-5}$, $k = 0.0295$, $Pr = 0.70$. Find h and loss per metre.

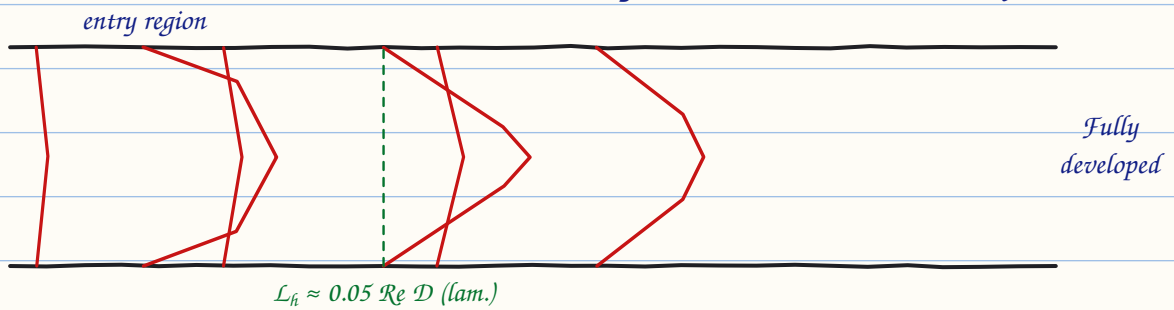
$Re_D = 10(0.025)/1.995\times10^{-5} = 12\,530$ → $C = 0.193$, $n = 0.618$. $Nu = 0.193$

$(12530)^{0.618}(0.7)^{1/3} = 0.193 \times 340 \times 0.888 = 58.3$

$h = 58.3 \times 0.0295/0.025 =$ **68.8 W/m²K**; $Q/L = h\,\pi\,D\,\Delta T = 68.8 \times \pi \times 0.025 \times 100 =$ **540 W/m**

Internal Flow – Convection in Pipes

Flow in a tube : $Re_D = \rho\, u_m\, D/\mu$. $Re < 2300$ laminar, 2300–4000 transition, $Re > 4000$ turbulent. For non-circular ducts use hydraulic diameter $D_h = 4A_c/P$.



→ Hydrodynamic entry length : laminar $L_h \approx 0.05\, Re\, D$; turbulent $L_h \approx 10\, D$. Thermal entry length : laminar $L_t \approx 0.05\, Re\, Pr\, D$

→ Bulk (mean) temperature T_b replaces T_∞: $Q = h\, A\, (T_s - T_b)$; energy balance $Q = \dot{m}\, c_p\, (T_{out} - T_{in})$

Case	Correlation (properties at T_b)	Range
Laminar, fully dev., T_s const	$Nu_D = 3.66$	$Re < 2300$
Laminar, fully dev., q const	$Nu_D = 4.36$	$Re < 2300$
Turbulent : Dittus–Boelter	$Nu_D = 0.023\, Re^{0.8}\, Pr^n$ (n=0.4 heating, 0.3 cooling)	$Re>10^4$, $0.6<Pr<160$, $L/D>10$
Sieder–Tate (viscous liquids)	$Nu_D = 0.027\, Re^{0.8}\, Pr^{1/3}\, (\mu_b/\mu_s)^{0.14}$	$Re>10^4$
Gnielinski (accurate)	$Nu = (f/8)(Re-1000)Pr \,/\, [1+12.7\sqrt{(f/8)}(Pr^{2/3}-1)]$	$3000<Re<5\times10^6$

Constant wall temperature (e.g. condensing steam outside):

$$(T_s - T_{out}) / (T_s - T_{in}) = \exp(-h\, P\, L / \dot{m}\, c_p)$$

$$Q = \dot{m}\, c_p\, (T_{out} - T_{in}) = h\, A\, \Delta T_{lm},\ \Delta T_{lm} = (\Delta T_{in} - \Delta T_{out}) / \ln(\Delta T_{in}/\Delta T_{out})$$

Constant heat flux (electric heater): $T_b(x) = T_{in} + q\, P\, x/(\dot{m}\, c_p)$ rises linearly; wall temperature $T_s = T_b + q/h$ also rises linearly after the entry length.

Solved Example 3 – Water heated in a tube

Water at 1 m/s through a 20 mm tube, 30°C in, 50°C out, wall at 80°C. Properties at 40°C: $\rho = 992$, $c_p = 4174$, $\nu = 0.658\times10^{-6}$, $k = 0.63$, $Pr = 4.3$. Find h and length.

$Re = 1(0.02)/0.658\times10^{-6} = 30\,400$ (turbulent). $Nu = 0.023\,(30400)^{0.8}(4.3)^{0.4} = 0.023 \times 3857 \times 1.79 = 159$

$h = 159(0.63)/0.02 = 5010\ W/m^2K$; $\dot{m} = 992(1)(\pi/4)(0.02)^2 = 0.3116\ kg/s$; $Q =$

$0.3116(4174)(20) = 26$ kW

$\ln[(80-50)/(80-30)] = -0.511 = -\,h\pi DL/\dot{m}c_p \rightarrow L = 0.511(1300.7)/(5010 \times 0.0628) = 2.11$ m

Pressure drop & pumping power (internal flow)

$$\Delta p = f\,(L/D)\,(\rho\, u_m^2/2)$$

$$f = 64/Re \text{ (laminar)}$$

$$f = 0.316\,Re^{-1/4} \text{ (turbulent smooth, } Re < 2\times10^4\text{)};\ f = 0.184\,Re^{-0.2} \text{ (} Re > 2\times10^4\text{)}$$

$$\text{Pumping power} = \dot{m}\,\Delta p/\rho = \mathcal{V}\Delta p$$

Example 3 data: $f = 0.316/(30400)^{0.25} = 0.0239$; $\Delta p = 0.0239\,(2.11/0.02)(992 \times 1^2/2) = 1.25$ kPa; power $= 0.3116 \times 1250/992 = 0.39$ W.

Solved Example 3B – Air in a rectangular duct

Duct 5 cm × 10 cm, air at 40°C, 5 m/s ($\nu = 1.70\times10^{-5}$, $k = 0.0265$, $Pr = 0.70$), being heated.

$D_h = 4A/P = 4(0.05 \times 0.10)/(2 \times 0.15) = 0.0667$ m; $Re = 5(0.0667)/1.70\times10^{-5} = 19\,600$ (turbulent)

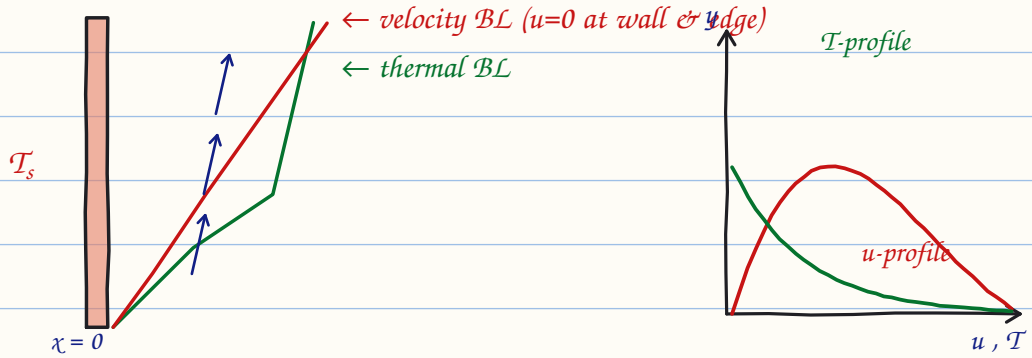
$Nu = 0.023\,(19600)^{0.8}(0.7)^{0.4} = 0.023 \times 2723 \times 0.867 = 54.3$; $h = 54.3(0.0265)/0.0667 = 21.6$ W/m²K

Solved Example 3C – Laminar flow of oil

Oil ($k = 0.14$ W/mK) at $Re = 500$ in a 10 mm tube with uniform wall heat flux. $Re < 2300 \rightarrow$ laminar, fully developed $Nu = 4.36 \Rightarrow h = 4.36(0.14)/0.01 = 61$ W/m²K (independent of velocity!). Thermal entry length $L_t = 0.05\,Re\,Pr\,D$ grows with Pr: for $Pr = 100$, $L_t = 0.05(500)(100)(0.01) = 25$ m.

Free (Natural) Convection

Fluid motion caused by buoyancy: heating reduces density, the lighter fluid near the wall rises. No external pump/fan; velocity is zero at the wall AND zero far away, with a maximum inside the boundary layer.



→ Boussinesq : ρ varies only in the buoyancy term; $\beta = -(1/\rho)(\partial\rho/\partial T)_p \rightarrow \beta = 1/T$ (K) for ideal gas

→ Governing parameter : $Gr = g\beta (T_s - T_\infty) L^3/\nu^2$, $Ra = Gr \cdot Pr$. Vertical plate: laminar $Ra < 10^9$, turbulent $Ra > 10^9$

→ Exact (similarity) solution, vertical plate: $Nu_x = 0.508\, Pr^{1/2} (0.952 + Pr)^{1/4}\, Gr_x^{1/4}$

Integral analysis of the free-convection BL (vertical plate)

Assume cubic-type profiles $T - T_\infty = (T_s - T_\infty)(1 - y/\delta)^2$ and $u = u_x (y/\delta)(1 - y/\delta)^2$.

Momentum balance (inertia = viscous + buoyancy):

$$d/dx \int_0^\delta u^2\, dy = -\nu\, (du/dy)_0 + g\beta \int_0^\delta (T - T_\infty)\, dy$$

Solving the momentum and energy integrals together gives (Squire–Eckert result):

$$\delta/x = 3.93\, Pr^{-1/2} (0.952 + Pr)^{1/4}\, Gr_x^{-1/4}$$

$$Nu_x = 0.508\, Pr^{1/2} (0.952 + Pr)^{1/4}\, Gr_x^{1/4}$$

→ $\delta \propto x^{1/4}$, $h_x \propto x^{-1/4}$ → average h over height L is $h_{avg} = (4/3)\, h_{x=L}$

Empirical correlations : $Nu = h L/k = C (Gr\, Pr)^n = C\, Ra^n$ (properties at T_f)

Geometry (characteristic L)	Range of Ra	C	n
Vertical plate / cylinder (height L)	$10^4 - 10^9$	0.59	1/4
Vertical plate / cylinder	$10^9 - 10^{13}$	0.10	1/3
Horizontal plate, hot face UP (L = A/P)	$10^4 - 10^7$	0.54	1/4
Horizontal plate, hot face UP	$10^7 - 10^{11}$	0.15	1/3
Horizontal plate, hot face DOWN	$10^5 - 10^{11}$	0.27	1/4
Horizontal cylinder (L = D)	$10^4 - 10^7$	0.48	1/4
Horizontal cylinder (L = D)	$10^7 - 10^{12}$	0.125	1/3
Sphere	$Ra < 10^{11}$	$Nu = 2 + 0.43\, Ra^{1/4}$	

Churchill–Chu (vertical, all Ra): $Nu = \{ 0.825 + 0.387\, Ra^{1/6} / [1 + (0.492/Pr)^{9/16}]^{8/27} \}^2$

Horizontal cylinder ($Ra_D < 10^{12}$): $Nu = \{ 0.6 + 0.387\, Ra_D^{1/6} / [1 + (0.559/Pr)^{9/16}]^{8/27} \}^2$

Note: A vertical cylinder behaves like a vertical plate if $D \geq 35\, L / Gr_L^{1/4}$. For a plate inclined at θ to the vertical replace g by $g \cos\theta$ ($\theta < 60°$).

Solved Example 4 – Vertical plate

Vertical plate 0.5 m high at 90°C in still air at 30°C ($T_f = 60°C$: $\nu = 1.896\times10^{-5}$, $k = 0.0287$, $Pr = 0.70$, $\beta = 1/333\ K^{-1}$). Find h and heat loss per metre width.

$Ra = g\beta\Delta T L^3 Pr/\nu^2 = 9.81(0.003)(60)(0.125)(0.7)/(3.595\times10^{-10}) = 4.3\times10^8 < 10^9 \rightarrow$ laminar

$Nu = 0.59\ (4.3\times10^8)^{1/4} = 0.59 \times 144 = 85$; $h = 85(0.0287)/0.5 =$ **4.9 W/m²K**; $Q = hL\Delta T = 4.9(0.5)(60) =$ **146 W/m**

Solved Example 5 – Horizontal pipe

Pipe $D = 0.1$ m at 80°C in air at 20°C ($T_f = 50°C$: $\nu = 1.79\times10^{-5}$, $k = 0.028$, $Pr = 0.703$, $\beta = 1/323$). $Ra_D = 9.81(1/323)(60)(0.001)(0.703)/(3.2\times10^{-10}) = 4.0\times10^6$

$Nu = 0.48\, Ra^{1/4} = 0.48(44.7) = 21.5$; $h = 21.5(0.028)/0.1 =$ **6.0 W/m²K**; $Q/L = h\,\pi\, D\,\Delta T = 6.0\,\pi\,(0.1)(60) =$ **113 W/m**

Solved Example 6 – Sphere in still air

Sphere $D = 0.2$ m at 100°C in air at 20°C ($T_f = 60°C$, $\nu = 1.896\times10^{-5}$, $k = 0.0287$, $Pr = 0.7$, $\beta = 1/333$). $Ra_D = 9.81(0.003)(80)(0.2)^3(0.7)/(3.595\times10^{-10}) = 3.7\times10^7$

$Nu = 2 + 0.43\, Ra^{1/4} = 2 + 0.43(77.8) = 35.5$; $h = 35.5(0.0287)/0.2 =$ **5.1 W/m²K**; $Q = h\,\pi D^2\,\Delta T = 5.1(0.1257)(80) =$ **51 W**

Which correlation? – quick decision guide

Situation	Use
Plate, flow along it	$Re_L < 5\times10^5$: $0.664Re^{1/2}Pr^{1/3}$; else $(0.037Re^{0.8}-871)Pr^{1/3}$
Cylinder, cross flow	Hilpert $Nu = C\,Re^n\,Pr^{1/3}$ or Churchill–Bernstein
Sphere, forced	Whitaker $Nu = 2 + (0.4Re^{1/2}+0.06Re^{2/3})Pr^{0.4}$
Pipe, laminar	$Nu = 3.66$ (T_s) or 4.36 (q); $Re<2300$
Pipe, turbulent	Dittus–Boelter / Sieder–Tate / Gnielinski
Free convection	$Nu = C\,Ra^n$; choose C,n by geometry & Ra
Mixed convection	$Gr/Re^2 \approx 1$: $Nu^3 = Nu_{forced}^{\,3} \pm Nu_{free}^{\,3}$

Remember: $Gr/Re^2 \gg 1$ → forced convection dominates ; $Gr/Re^2 \ll 1$ → free convection dominates ; ≈ 1 → mixed. Properties of fluid: evaluate at T_f; in pipes at bulk mean temperature.

Solved Example 7 – Plate with mixed boundary layer

Air at 20°C, $U = 30$ m/s flows over a 2 m long plate at 60°C ($T_f = 40$°C: $\nu = 1.70\times10^{-5}$, $k = 0.0265$, $Pr = 0.70$). Find h_{avg} and heat loss per metre width.

$Re_L = 30(2)/1.70\times10^{-5} = 3.53\times10^6 > 5\times10^5$; transition at $x_c = 5\times10^5(1.7\times10^{-5})/30 = 0.28$ m → mixed BL

$Nu_L = (0.037\,Re^{0.8} - 871)Pr^{1/3} = (0.037 \times 1.73\times10^5 - 871)(0.888) = 4915$; $h = 4915(0.0265)/2 =$ **65 W/m²K**

$Q = hL\Delta T(1\text{ m}) = 65(2)(40) =$ **5.2 kW per metre width**

Solved Example 8 – Hot horizontal plate facing up

Square plate 0.3 m × 0.3 m at 80°C in air at 20°C ($T_f = 50$°C; $\nu = 1.79\times10^{-5}$, $k = 0.028$, $Pr = 0.703$, $\beta = 1/323$). $L_c = A/P = 0.09/1.2 = 0.075$ m.

$Ra = 9.81(1/323)(60)(0.075)^3(0.703)/(3.2\times10^{-10}) = 1.7\times10^6$; $Nu = 0.54\,Ra^{1/4} = 0.54(36.0) = 19.5$; $h = 19.5(0.028)/0.075 = 7.3$

$Q_{top} = 7.3(0.09)(60) =$ **39 W**. (If the hot face were DOWN: $Nu = 0.27\,Ra^{1/4}$ → half the loss.)

Laminar vs Turbulent – comparison

Aspect	Laminar	Turbulent
Flow pattern	smooth layers	eddies, mixing
BL thickness δ	$\propto x^{1/2}$	$\propto x^{4/5}$
$Nu_x \propto$	$Re^{1/2}$	$Re^{0.8}$
h, C_f	low	high
Where	$Re_x < 5\times10^5$ / $Re_D < 2300$	$Re_x > 5\times10^5$ / $Re_D > 4000$

Properties of water (approx.)

T (°C)	ρ (kg/m^3)	c_p (J/kgK)	k (W/mK)	ν (m^2/s)	Pr
20	998	4182	0.598	1.00×10^{-6}	7.0
40	992	4179	0.631	0.66×10^{-6}	4.3
60	983	4185	0.654	0.48×10^{-6}	3.0
80	972	4197	0.670	0.36×10^{-6}	2.2

Important Questions (exam oriented)

1. Distinguish free and forced convection. Define h, Nu, Pr, Gr and Ra with physical significance.
2. Explain hydrodynamic and thermal boundary layers on a flat plate with neat sketches.
3. Obtain δ, C_f and Nu_x for laminar flow over a flat plate using the integral method.
4. Explain Reynolds–Colburn analogy and its limitations.
5. Explain development of the boundary layer on a heated vertical plate. State correlations for plates & cylinders.
6. Write Dittus–Boelter equation. Explain thermal and hydrodynamic entry lengths in pipes.

Quick viva (short answers)

Q1.Why is h not a property? – it depends on flow, geometry and fluid, unlike k.

Q2.Why is the gas-side heat transfer coefficient low? – low k and ρ of gas → fins on gas side.

Q3.Pr of liquid metals? – □ 1 (0.01–0.03): thermal BL thicker than velocity BL.

Q4.Why does turbulence raise h? – eddy mixing carries energy across the layer, thinning

the sub-layer.

Q5.When does the flat-plate BL become turbulent? – $Re_x \approx 5\times10^5$.

Q6.Which temperature is used for properties? – film temperature; bulk mean temperature for pipes.

Q7.What is the role of Gr in natural convection? – it replaces Re (buoyancy/viscous).

